# Dr Fan Zhang

## Contact Information

Honda Research Institute EU
Frankfurt, Germany

Email: fanzhang0@outlook.com
www: fan6zh.github.io

## Research Interests

Robot Manipulation, Sim-to-Real Learning, Self-Supervised Learning, World Models, Flow Matching

## Professional Appointments

**Senior Scientist**, 2025-present
**Guest Scientist**, 2024
Honda Research Institute EU

- *Note: Secured AI Research Scientist role at Meta FAIR in 2025; impacted following company-wide restructuring.*

**Eric and Wendy Schmidt AI in Science Postdoctoral Fellow**, Schmidt Futures, 2023-2024
**Research Associate**, 2021-2023
Imperial College London (UK Global Talent Visa, sponsored by Royal Academy of Engineering)

## Education

**Ph.D. in Electrical and Electronic Engineering (Robotics)**, 2016-2020
Imperial College London
Thesis**:** Perception and Manipulation in Robotic-Assisted Dressing
Supervisor: Prof. Yiannis Demiris (Royal Academy of Engineering Chair in Emerging Technologies)

## Awards

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|---|---|
| Outstanding Paper Award, IROS 2025 Workshop on The Art of Robustness | 2025 |
| Best Research Paper, AI & Robotics Research Awards, held by UKRI TAS Hub and RAI | 2025 |
| The UK Best PhD in Robotics Award 1st place, held by Advanced Robotics @ Queen Mary | 2020 |
| Best Student Paper Award, IEEE International Conference on Mechatronics and Automation | 2016 |

## Highlighted Publications

**Learning Robot Manipulation from Audio World Models**
Fan Zhang, Michael Gienger
arXiv, 2025. (paper)

**Affordance-based Manipulation with Flow Matching**
Fan Zhang, Michael Gienger
IEEE Transactions on Robotics (conditionally accepted), 2026. (paper, website, code 235★)

**Learning Garment Manipulation Policies towards Robot-Assisted Dressing**
Fan Zhang, Yiannis Demiris
Science Robotics, 2022. (paper, website, code)

**Probabilistic Real-Time User Posture Tracking for Personalized Robot-Assisted Dressing**
Fan Zhang, Antone Cully, Yiannis Demiris
IEEE Transactions on Robotics, 2019. (paper, website, code)

## Talks

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|---|---|
| Intelligent Systems and Networks (ISN) Group Seminar, Imperial College London | 2025 |
| IROS Workshop on Manipulation of Deformable Objects (RoMaDo) | 2025 |
| IPAB visitor seminar, Edinburgh Centre for Robotics | 2025 |
| YorRobots Seminar, University of York | 2025 |
| Autonomous Agents Research Group, The University of Edinburgh | 2025 |
| Adaptive and Intelligent Robotics Lab, Imperial College London | 2025 |
| Guest Lecture, TAMS, University of Hamburg | 2024 |
| Social AI & Robotics Laboratory, King's College London | 2023 |
| Tsinghua University, (live audience: 150,000) | 2022 |
| Apple | 2022 |
| Human Motion Analysis for Healthcare Applications, IET (video) | 2019 |
| The Hamlyn Centre, Imperial College London | 2017 |

## In the Press

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|---|---|
| Robotic nurse can dress a mannequin in a hospital gown, New Scientist | 2022 |
| Robotic nurse that helps you dress could aid staff shortage, Bloomberg | 2019 |
| The Times, Daily Mail, Telegraph, TechXplore | 2019 |

## Academic Activities

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| Inaugural Meeting - Centre for AI in Assistive Autonomy | Invited Guest |
| IROS 2025 Session: Factory Automation and Failure Detection | Co-chair |
| RSS 2024 Workshop: Learning for Assistive Robotics | Organizer |
| Imperial-X Breaking Topics in AI conference, Schmidt Futures | Organizer |
| ICRA 2023 Workshop: Emerging Paradigms for Assistive Robotic Manipulation | Organizer |
| Frontiers in Robotics and AI-Robot Learning and Evolution | Review Editor |
| Scientific Reports, T-RO, IJRR, ICRA, IROS, RA-L, RSS, WACV, Humanoids | Reviewer |
| The Centre for AI in Assistive Autonomy, The University of Edinburgh | Collaborator |
| Doctoral annual review (Leonard Hinckeldey), The University of Edinburgh | Committee |

## Research Mentorship

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|---|---|
| Jonathan Sutphen (Msc, TU Darmstadt, Robot Manipulation with Audio) | 2025-2026 |
| Leonard Hinckeldey (PhD, The University of Edinburgh, Multi-Agent RL) | 2025-present |
| Amirreza Razmjoo (PhD, EPFL, Robot Failure Recovery) | 2024-2025 |
| Jenny Fu (PhD, Cornell University, Robot Character Learning) | 2024-2025 |
| Nikki Zhong (PhD, Imperial college London, Human Motion Modeling) | 2022-2024 |

## Research Experience

---- **General Multi-Modal Robot Manipulation**

- Multimodal (audio-vision) world model for learning robot manipulation with latent flow matching.
- Parameter-efficient prompt tuning for learning manipulation affordance with VLM.
- Generative learning for robot manipulation with special focus on flow matching.
- Applications: robot failure recovery with compositional diffusion models, robotic expressions and characters learning with flow matching, piano playing with audio world models.
- The above works have been published in: IEEE Transactions on Robotics (conditionally accepted), IROS, workshops at CoRL and ROMAN, and received Outstanding Paper Award at IROS 2025 Workshop on The Art of Robustness.
- Developed an open-source robot manipulation repository, achieving 235+ stars on GitHub (code).

---- **Robot-Assisted Dressing for Bedridden Patients**

- Deformable visual-tactile manipulation for garment grasping unfolding using deep Q learning from demonstration and model-based reinforcement learning.
- Sim-to-real garment physics learning and robot manipulation policy transfer in physics domain using self-supervised learning.
- Contrastive learning for human motion modeling, including personalized user impairments modeling and dancing performance assessment.
- Building large real and synthetic dataset of garment, including RGB-D, event images and pointcloud.
- Real-time user posture tracking using vision and haptic information with a probabilistic particle filter.
- Building controllers for hierarchical multi-task human-robot interaction.
- The above works have been published in top journals and conferences: Science Robotics, IEEE Transactions on Robotics, RA-L, ICRA, IROS, ICASSP, ICMA (Best Student Paper). Live demos for National Health Service, Apple, etc.
- Media covered by Bloomberg, The Times, Daily Mail, Telegraph , NewScientist, TechXplore.
- The Science Robotics paper has been selected as the Best Research Paper, Early Career Researcher, 2025 AI & Robotics Research Awards, held by UKRI TAS Hub and Responsible Ai UK.
- Based on above works, I have been awarded the UK Best PhD in Robotics Award 2020 1st place, held by the Centre for Advanced Robotics @ Queen Mary (ARQ).
- The above works are supported by MURI UKRI Closed-Loop Multisensory Brain-Computer Interface for Enhanced Decision, TAS Trust Node UKRI Trustworthy Autonomous Systems Node in Trust, Innovate UK D-RISK Learning Edge Cases for Autonomous Vehicles.

## Other Publications

1) **Assistax: A Hardware-Accelerated Reinforcement Learning Benchmark for Assistive Robotics**
   Leonard Hinckeldey ~Leonard_Hinckeldey1 , Elliot Fosong, Rimvydas Rubavicius, Elle Miller, Trevor McInroe, Fan Zhang, Patricia Wollstadt, Stefano V. Albrecht, Subramanian Ramamoorthy
   arXiv, 2026. (paper)

2) **CARMA: Context-Aware Situational Grounding of Human-Robot Group Interactions by Combining Vision-Language Models with Object and Action Recognition**
   Joerg Deigmoeller, Stephan Hasler, Nakul Agarwal, Daniel Tanneberg, Anna Belardinelli, Reza Ghoddoosian, Chao Wang, Felix Ocker, Fan Zhang, Behzad Dariush, Michael Gienger
   ICRA, 2026. (paper)

3) **Composition of Conditional Diffusion Policies with Guided Sampling**
   Amirreza Razmjoo, Sylvain Calinon, Michael Gienger, Fan Zhang
   IROS, 2025. (paper, website)

4) **Generation of Real-time Robotic Emotional Expressions Learning from Human Demonstration in Mixed Reality**
   Chao Wang, Michael Gienger, Fan Zhang
   IROS 2025 Foundation Models for Robotic Design workshop. (paper, website)

5) **Robot manipulation with flow matching**
   Fan Zhang, Michael Gienger
   CoRL 2024 Workshop on Mastering Robot Manipulation in a World of Abundant Data.

6) **Contrastive Self-Supervised Learning for Automated Multi-Modal Dance Performance Assessment**
   Yun Zhong, Fan Zhang, Yiannis Demiris
   ICASSP, 2023. (paper, video, code)

7) **Visual-Tactile Learning of Garment Unfolding for Robot-Assisted Dressing**
   Fan Zhang, Yiannis Demiris

RA-L, 2023. (paper, video)

8) **Grasp-oriented fine-grained cloth segmentation without real supervision**
Ruijie Ren, Mohit Gurnani Rajesh, Jordi Sanchez-Riera, Adrian Lopez-Rodriguez, Fan Zhang, Yurun Tian, Guillem Alenyà, Antonio Agudo, Yiannis Demiris, Krystian Mikolajczyk, Francesc Moreno Noguer
6th International Conference on Machine Vision and Applications, 2023.

9) **Learning Grasping Points for Garment Manipulation in Robot-Assisted Dressing**
Fan Zhang, Yiannis Demiris
ICRA, 2020. (paper, video)

10) **Preoperative optimization of the surgical robot considering internal diversity of workspace**
Zhiyuan Yan, Zhijiang Du, Fan Zhang, Weidong Wang
Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science, 2018.

11) **Personalized Robot-Assisted Dressing using User Modeling in Latent Spaces**
Fan Zhang, Antone Cully, Yiannis Demiris
IROS, 2017. (paper, video)

12) **Preoperative Planning for the Multi-Arm Surgical Robot using PSO-GP-based Performance Optimization**
Fan Zhang, Zhiyan Yuan, Zhijiang Du
ICRA, 2017. (paper, video)

13) **Preoperative Setup Planning for Robotic Surgery Based on a Simulation Platform and Gaussian Process**
Fan Zhang, Zhiyan Yuan, Zhijiang Du.
IEEE International Conference on Mechatronics and Automation (ICMA), 2016. (paper)

14) **An under-actuated manipulation controller based on Workspace Analysis and Gaussian Processes**
Fan Zhang, Yanyu Su, Xiang Zhang, Wei Dong, Zhijiang Du
IROS, 2015. (paper, video)